

Speciation of methyl- and ethyl-mercury in hair of breastfed infants acutely exposed to thimerosal-containing vaccines.



BACKGROUND: Different chemical forms of mercury occur naturally in human milk. The most controversial aspect of early post-natal exposure to organic mercury is ethylmercury (EtHg) in thimerosal-containing vaccines (TCV) still being used in many countries. Thus exclusively breastfed infants can be exposed to both, fish derived methylmercury (MeHg) in maternal diets and to EtHg from TCV. The aim of the study is to evaluate a new analytical method for ethyl and methyl mercury in hair samples of breastfed infants who had received the recommended schedule of TCV. **METHODS:** The hair of infants (<12 months) that had been exposed to TCV (Hepatitis B and DTaP) was analysed. A method coupling isothermal gas chromatography with cold-vapor atomic fluorescence spectrometry was used for MeHg which can also speciate EtHg in biological matrices. **RESULTS:** In 20 samples of infants' hair, all but two samples showed variable amounts of MeHg (10.3 to 668 ng/g), while precise and reliable concentrations of EtHg (3.7 to 65.0 ng/g) were found in 15 of the 20 samples. A statistically significant inverse association ($r=-0.5572$; $p=0.0384$) was found between hair-EtHg concentrations and the time elapsed after the last TCV shot. **CONCLUSIONS:** The analytical method proved sensitive enough to quantify EtHg in babies' hair after acute exposure to thimerosal in vaccine shots. Provided that the mass of hair was above 10mg, organic-mercury exposure during early life can be speciated, and quantified in babies' first hair, thus opening opportunities for clinical and forensic studies. Copyright © 2011 Elsevier B.V. All rights reserved.